GPU-Accelerated Optimization for Motion Planning with NVIDIA CuRobo for Industrial Robotics

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Overview

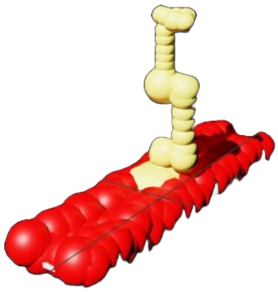
In modern industrial automation, efficient motion planning remains a critical challenge, particularly for **multi-axis robotic systems** operating in dynamic environments. Traditional **CPU-based motion planners struggle** with real-time responsiveness, often leading to suboptimal trajectories and inefficient collision avoidance.

Low-Cost GPU Platform

Our project addresses this issue by leveraging GPU-accelerated motion planning with NVIDIA's CuRobo library on an **Nvidia Jetson Nano**. This approach enables **62x faster trajectory generation** compared to conventional motion planners, ensuring real-time collision avoidance. Through MuJoCo simulation, ROS2 integration, and seventh-axis gantry expansion, we demonstrated major improvements, making GPU acceleration a game-changer for industrial robotics.

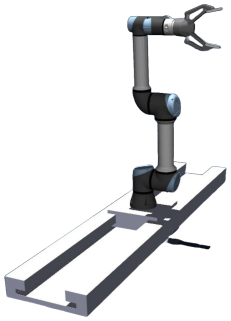
Self-Collision Representation

The robot and any additional joints are represented as **spheres** to define **self-collision**. This defines the number of joints in the plan.



Motion Planning Execution

The motion planner **generates and publishes trajectories**, which are then fed into the simulation.
This ensures **accurate, real-time motion execution** while adapting to dynamic environments.



Apply Trajectories

Update World

Obstacle Representation

The surrounding obstacles are defined using **simple geometries (cubes, spheres, etc.), meshes, or point cloud data**.
This allows the planner to **efficiently detect and avoid obstacles** during motion generation.

Mujoco Simulation Integration

The Mujoco simulation loop continuously updates obstacle positions based on physical contact models. Any obstacles affecting motion are updated in real-time within the CuRobo motion planner.